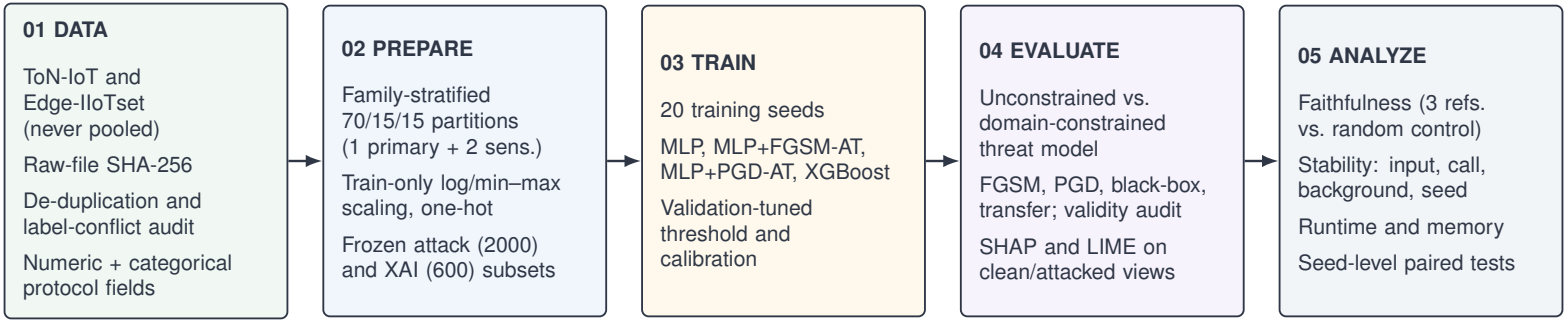
01 DATA
ToN-IoT and Edge-IIoTset (never pooled)
Raw-file SHA-256
De-duplication and label-conflict audit
Numeric + categorical protocol fields
02 PREPARE
Family-stratified 70/15/15 partitions (1 primary + 2 sens.)
Train-only log/min–max scaling, one-hot
Frozen attack (2000) and XAI (600) subsets
03 TRAIN
20 training seeds
MLP, MLP+FGSM-AT, MLP+PGD-AT, XGBoost
Validation-tuned threshold and calibration
04 EVALUATE
Unconstrained vs. domain-constrained threat model
FGSM, PGD, black-box, transfer; validity audit
SHAP and LIME on clean/attacked views
05 ANALYZE
Faithfulness (3 refs. vs. random control)
Stability: input, call, background, seed
Runtime and memory
Seed-level paired tests